A finite maximal universal tree forces a Sigma^0_1 formula to have a last true instance

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Editorial note. The counted unit is the article's own Lemma lem:sem-wf-finite-tree-forcing at source lines 893-897 together with the proof environment that begins on the next line (898-912); the pairing is the ordinary adjacent one.

Editorial note. The retained context at source lines 819-829 and 853-861 is the author's own definitions of nodes of a condition, of validity, of trees and of universal trees; the retained context at source lines 841-844 and 880-882 is the statement of the two lemmas that the counted proof invokes as its first two steps, both counted by sibling cards of this lane. They are retained as prerequisites so that the author Cref calls resolve inside the retained material and no navigation stub is needed.

Editorial note. The retained preamble reproduces verbatim the author's document class at source line 2, his theorem declarations at source lines 16-25, his macro definitions at source lines 46-88 and 91, and his natural-operation macros at source lines 104-107, and it retains mathabx and cleveref; the stix-dependent LS2 block and the inputenc load with the utf8x option are not retained, the former because the host lacks stix and the latter because it fights xelatex. Nothing was renamed or redefined.

Editorial note. No retained line contains a citation or a cross-reference to anything outside the retained material, so the card carries no bibliography and needs no external navigation pointer.

Editorial note. The article's conservation theorems, its COH and WKL lemmas and the other lemmas of the SEM section are not counted here and are not used as premises.

Allowed standard prerequisites (editorial).

- Nodes, validity, trees and universal trees of the SEM forcing. A node is an M-finite sequence of pairs of a bit and an M-finite R-transitive subset of X_a; it is valid when each block lies in the corresponding side of the colouring; a tree is a downward closed set of nodes and it is universal when every non-leaf node can be extended on one of the two sides of every finite partition of a sufficiently long initial segment; the author's own definitions

are retained at source lines 819-829 and 853-861.. Fixes all the objects of the counted lemma, namely the condition, the Sigma^0_1 formula, the finite maximal universal tree and the notion of a node satisfying the formula.

- Valid leaves and the extension they produce. Every M-finite universal tree has a valid leaf, and a valid node yields an extension of the underlying condition; both statements are retained with this card at source lines 880-882 and 841-844.. The proof starts from a valid leaf of the tree and takes the extension that the leaf produces, so both statements are used as the first two steps.

- The family of sets X_a, compactness and the class of partitions. The conditions are built from the family of sets X_a indexed by integers, with G almost contained in X_b for the relevant b; the class of partitions of X_b satisfying the property obtained from maximality is a non-empty Pi^0_1(Y plus X_b) class, and by definition of the family there are indices c and d partitioning X_b inside that class.. The proof needs compactness to pass from the partitions of every initial segment to a partition of the whole X_b, and then the family of sets to realise that partition by two of its members.

Definition 0.1. *Fix a condition* (σ, a).

(1) *A* (σ, a)-node *is an M-finite sequence of the form*

$$\nu = \langle (i_0, \rho_0), (i_1, \rho_1), \dots (i_{q-1}, \rho_{q-1}) \rangle$$

such that for every $s < q$, $i_s < 2$, $\rho_s \subseteq X_a$ *is a non-empty M-finite R-transitive set with* $\min \rho_s > \max \rho_{s-1}$ *(letting* $\rho_{-1} = \sigma$*), and for every* $s < t < q$, *if* $i_s = 0$ *then* $\rho_s \to_R \rho_t$, *and if* $i_s = 1$ *then* $\rho_t \to_R \rho_s$. *We then write*

$$\pi(\nu) = \sigma \cup \rho_0 \cup \dots \cup \rho_{q-1}$$

(2) *This* (σ, a)*-node is* valid *if for every* $s < q$, $\rho_s \subseteq A_{i_s}(R)$

(3) *A* (σ, a)*-node* ν satisfies *a* $\Sigma^0_1(\mathcal{M})$*-formula* $\varphi(G, x) \equiv \exists i \psi(G\restriction i, x)$ *if* $\psi(\pi(\nu), |\nu| - 1)$ *holds.*

Lemma 0.2. *Fix a valid* (σ, a)*-node* ν. *Then* $(\pi(\nu), b)$ *is an extension of* (σ, a) *for some* $b \in M$.

Definition 0.3. *Fix a condition* (σ, a).

(1) *A* (σ, a)-tree *is a set of* (σ, a)*-nodes closed downard under the prefix relation.*

(2) *A (σ, a)-tree E is* universal *if for every (σ, a)-node ν which is not a leaf, there is some $r \in M$ such that for every M-finite 2-partition $A_0 \sqcup A_1 = \{0, \dots, r-1\}$, there is a side $i < 2$ and an M-finite R-transitive $\rho \subseteq A_i \cap X_a$ such that $\nu \cdot \langle i, \rho \rangle \in E$.*

(3) *A (σ, a)-tree E satisfies a $\Sigma^0_1(\mathcal{M})$ formula if all its nodes satisfies it.*

Lemma 0.4. *Every M-finite universal (σ, a)-tree E has a valid leaf.*

Lemma 0.5. *Let (σ, a) be a condition, $\varphi(G, x)$ be a $\Sigma^0_1(\mathcal{M})$ formula, and E be an M-finite maximal universal (σ, a)-tree satisfying $\varphi(G, x)$. Then there is an extension $(\tau, c) \leqslant (\sigma, a)$ and some $q \in M$ such that $(\tau, c) \Vdash \varphi(G, q-1)$ and $(\tau, c) \Vdash \neg\varphi(G, q)$.*

Proof. By Lemma 0.4, there is a valid leaf $\nu \in E$. Since E satisfies $\varphi(G, x)$, $\varphi(\pi(\nu), |\nu| - 1)$ holds. By Lemma 0.2, there is some $b \in M$ such that $(\pi(\nu), b) \leqslant (\sigma, a)$.

By maximality of the tree, for every $s \in M$, there exists a 2-partition $A_0 \sqcup A_1 = \{0, \dots, s\}$, such that for every side $i < 2$ and every M-finite $\rho \subseteq A_i \cap X_a$ such that $\nu \cdot \langle i, \rho \rangle$ is a (σ, a)-node, then $\varphi(\pi(\nu) \cup \rho, |\nu|)$ does not hold.

Note that by choice of X_b, for every M-finite R-transitive $\rho \subseteq A_i \cap X_b$, $\nu \cdot \langle i, \rho \rangle$ is a (σ, a)-node. Thus, for every $s \in M$, there exists a 2-partition $A_0 \sqcup A_1 = \{0, \dots, s\}$, such that the following property holds:

> (P): For every side $i < 2$ and every M-finite R-transitive $\rho \subseteq A_i \cap X_b$, $\varphi(\pi(\nu) \cup \rho, |\nu|)$ does not hold.

By compactness, there exists a 2-partition $A_0 \sqcup A_1 = X_b$ such that (P) holds. Moreover, the class $\mathcal{C}$ of all such partitions is $\Pi^0_1(Y \oplus X_b)$ and non-empty. By definition of $\vec{X}$, there exists some $c, d \in M$ such that X_c, X_d is a 2-partition of X_b in $\mathcal{C}$. Moreover, since $G \subseteq^* X_b$, either $G \subseteq^* X_c$ or $G \subseteq^* X_d$. Since they play a symmetric role, say $G \subseteq^* X_c$. Then (τ, c) is an extension of (τ, b) forcing $\varphi(G, |\nu| - 1)$ and $\neg\varphi(G, |\nu|)$. □